

Statement of Work
National Oceanic and Atmospheric Administration (NOAA)
National Marine Fisheries Service (NMFS)
Center for Independent Experts (CIE) Program
External Independent Peer Review

Alaska Sablefish Assessment

Background

The National Marine Fisheries Service (NMFS) is mandated by the Magnuson-Stevens Fishery Conservation and Management Act, Endangered Species Act, and Marine Mammal Protection Act to conserve, protect, and manage our nation's marine living resources based upon the best scientific information available (BSIA). NMFS science products, including scientific advice, are often controversial and may require timely scientific peer reviews that are strictly independent of all outside influences. A formal external process for independent expert reviews of the agency's scientific products and programs ensures their credibility. Therefore, external scientific peer reviews have been and continue to be essential to strengthening scientific quality assurance for fishery conservation and management actions.

Scientific peer review is defined as the organized review process where one or more qualified experts review scientific information to ensure quality and credibility. These expert(s) must conduct their peer review impartially, objectively, and without conflicts of interest. Each reviewer must also be independent from the development of the science, without influence from any position that the agency or constituent groups may have. Furthermore, the Office of Management and Budget (OMB), authorized by the Information Quality Act, requires all federal agencies to conduct peer reviews of highly influential and controversial science before dissemination, and that peer reviewers must be deemed qualified based on the OMB Peer Review Bulletin standards.

(http://www.cio.noaa.gov/services_programs/pdfs/OMB_Peer_Review_Bulletin_m05-03.pdf).

Further information on the CIE program may be obtained from www.ciereviews.org.

Scope

Potential changes to the Alaska sablefish assessment have been proposed. These changes include development of a new fishery catch per unit effort (CPUE) index, incorporation of estimates of whale depredation, and alternatives to the methods for apportionment of catch by area. These changes could have a significant impact on the assessment and on stakeholders. The authors request a review of these potential new changes to the assessment and guidance on best practices for implementation. The Terms of Reference (TORs) of the peer review and the tentative agenda of the meeting are below.

Requirements

NMFS requires three reviewers to conduct an impartial and independent peer review in accordance with the SOW, OMB Guidelines, and the TORs below. The reviewers shall have working knowledge and recent experience in the application of 1) Stock assessment/Population Dynamics; 2) Generalized Linear Mixed Modeling/Generalized Additive Modeling/Generalized Linear Modeling; 3) Fisheries Management, and 4) Spatially-explicit assessment modeling

Tasks for reviewers

- Review the following background materials and reports prior to the review meeting:

NOAA. 2009. Independent review reports, N. Klaer, M. Armstrong, and J. Casey.

https://www.st.nmfs.noaa.gov/Assets/Quality-Assurance/documents/peer-review-reports/2009/2009_04_02%20Armstrong%20Alaska%20sablefish%20assessment%20review%20report.pdf

https://www.st.nmfs.noaa.gov/Assets/Quality-Assurance/documents/peer-review-reports/2009/2009_04_02%20Klaer%20Alaska%20sablefish%20assessment%20review%20report.pdf

https://www.st.nmfs.noaa.gov/Assets/Quality-Assurance/documents/peer-review-reports/2009/2009_04_02%20Casey%20Alaska%20sablefish%20assessment%20review%20report.pdf

Fenske, K., D.H. Hanselman, and T.J. Quinn II. *In prep.* A spatial assessment model for Alaska sablefish and the implications for the apportionment strategy.

Hanselman, D.H., C. Lunsford, and C. Rodgveller. 2009. Appendix 3C. Responses to CIE recommendations for the Alaska sablefish assessment. *In* Stock assessment and fishery evaluation report for the groundfish resources of the GOA and BS/AI. North Pacific Fishery Management Council, 605 W 4th Ave, Suite 306 Anchorage, AK 99501.

Hanselman, D.H., C. Lunsford, C. Rodgveller, and M. Peterson. 2014. Appendix 3C. Alaska sablefish research update. *In* Stock assessment and fishery evaluation report for the groundfish resources of the GOA and BS/AI. North Pacific Fishery Management Council, 605 W 4th Ave, Suite 306 Anchorage, AK 99501.

Hanselman, D.H., C. Lunsford, and C. Rodgveller. 2015. Assessment of the sablefish stock in Alaska. *In* Stock assessment and fishery evaluation report for the groundfish resources of the GOA and BS/AI. North Pacific Fishery Management Council, 605 W 4th Ave, Suite 306 Anchorage, AK 99501.

Hanselman, D.H., B. Pyper, and M. Peterson. *In prep.* Effects and implications of sperm whale depredation on longline surveys for Alaskan sablefish.

NPFMC 2015. Minutes from the November Groundfish Plan Team and the December Scientific and Statistical Committee relevant to sablefish.

Peterson, M.J. and D.H. Hanselman. *In prep.* Estimation of the relative and absolute impacts of whale depredation on the Alaska longline fishery.

Hanselman, D.H., J. Heifetz, K.B. Echave, and S.C. Dressel. 2015. Move it or lose it: Movement and mortality of sablefish tagged in Alaska. *Can. J. Fish. Aquat. Sci.* 72(2): 238-25.

Heifetz, J., J. T. Fujioka, and T. J. Quinn II. 1997. Geographic apportionment of sablefish, *Anoplopoma fimbria*, harvest in the northeastern Pacific Ocean. *In* M. Saunders and M. Wilkins (eds.). Proceedings of the International Symposium on the Biology and Management of Sablefish. pp 229-238. NOAA Tech. Rep. 130.

Peterson, M.J., F. Mueter, D.H. Hanselman, C.R. Lunsford, C. Matkin, and H. Fearnbach. 2013. Killer whale (*Orcinus orca*) depredation effects on catch rates of six groundfish species: Implications for commercial longline fisheries in Alaska. *ICES J. Mar. Sci.* 70: 1220-1232.

Shotwell, S. K., D. H. Hanselman, and I. M. Belkin. 2014. Toward biophysical synergy: investigating advection along the Polar Front to identify factors influencing Alaska Sablefish recruitment. *Deep-Sea Research Part II Topical Studies in Oceanography* 107:40–53

- Attend and participate in the panel review meeting
- After the review meeting, reviewers shall conduct an independent peer review in accordance with the requirements specified in this SOW, OMB guidelines, and TORs, in adherence with the required formatting and content guidelines; reviewers are not required to reach a consensus
- Each reviewer may assist the Chair of the meeting with contributions to the summary report, if required by the TORs
- Deliver their reports to the Government according to the specified milestone dates

Foreign National Security Clearance

When reviewers participate during a panel review meeting at a government facility, the NMFS Project Contact is responsible for obtaining the Foreign National Security Clearance approval for reviewers who are non-US citizens. For this reason, the reviewers shall provide requested information (e.g., first and last name, contact information, gender, birth date, passport number, country of passport, travel dates, country of citizenship, country of current residence, and home country) to the NMFS Project Contact for the purpose of their security clearance, and this information shall be submitted at least 30 days before the peer review in accordance with the NOAA Deemed Export Technology Control Program NAO 207-12 regulations available at the Deemed Exports NAO website: <http://deemedexports.noaa.gov/> and http://deemedexports.noaa.gov/compliance_access_control_procedures/noaa-foreign-national-registration-system.html. The contractor is required to use all appropriate methods to safeguard Personally Identifiable Information (PII).

Place of Performance

The place of performance shall be at the contractor's facilities, and at the NOAA Fisheries Alaska Fisheries Science Center in Juneau, Alaska.

Period of Performance

The period of performance shall be from the time of award through June 30, 2016. Each reviewer's duties shall not exceed 14 days to complete all required tasks.

Schedule of Milestones and Deliverables: The contractor shall complete the tasks and deliverables in accordance with the following schedule.

Within two weeks of award	Contractor selects and confirms reviewers
No later than April 26, 2016	Contractor provides the pre-review documents to the reviewers
May 10-12, 2016	Panel review meeting
May 27, 2016	Contractor receives draft reports
June 10, 2016	Contractor submits final reports to the Government

Applicable Performance Standards

The acceptance of the contract deliverables shall be based on three performance standards:
(1) The reports shall be completed in accordance with the required formatting and content (2)
The reports shall address each TOR as specified (3) The reports shall be delivered as specified in
the schedule of milestones and deliverables.

Travel

All travel expenses shall be reimbursable in accordance with Federal Travel Regulations (<http://www.gsa.gov/portal/content/104790>). International travel is authorized for this contract. Travel is not to exceed \$23,000.

Restricted or Limited Use of Data

The contractors may be required to sign and adhere to a non-disclosure agreement.

Peer Review Report Requirements

1. The report must be prefaced with an Executive Summary providing a concise summary of the findings and recommendations, and specify whether or not the science reviewed is the best scientific information available.
2. The report must contain a background section, description of the individual reviewers' roles in the review activities, summary of findings for each TOR in which the weaknesses and strengths are described, and conclusions and recommendations in accordance with the TORs.
 - a. Reviewers must describe in their own words the review activities completed during the panel review meeting, including a brief summary of findings, of the science, conclusions, and recommendations.
 - b. Reviewers should discuss their independent views on each TOR even if these were consistent with those of other panelists, but especially where there were divergent views.
 - c. Reviewers should elaborate on any points raised in the summary report that they believe might require further clarification.
 - d. Reviewers shall provide a critique of the NMFS review process, including suggestions for improvements of both process and products.
 - e. The report shall be a stand-alone document for others to understand the weaknesses and strengths of the science reviewed, regardless of whether or not they read the summary report. The report shall represent the peer review of each TOR, and shall not simply repeat the contents of the summary report.
3. The report shall include the following appendices:
 - Appendix 1: Bibliography of materials provided for review
 - Appendix 2: A copy of this Statement of Work
 - Appendix 3: Panel membership or other pertinent information from the panel review meeting.

Terms of Reference for the Peer Review

Alaska Sablefish Assessment

- a. Evaluation, findings, and recommendations on quality of input data and methods used to process them for inclusion in the assessment.
- b. Evaluation, findings, and recommendations of the analytical approach used to assess stock condition and stock status.
- c. Evaluation, findings, recommendations on estimation and strategies for accounting for whale depredation
 - a. Are the data and methods used in estimating depredation effects sufficient?
 - b. Should depredation estimates be used in the assessment model, and if so, how?
- d. Evaluation, findings, recommendations of areal harvest apportionment strategy as related to movement and optimizing spawning stock biomass
 - a. Are there biological reasons to adjust apportionment by area?
 - b. Is stability more important than close alignment to annual areal abundance changes?
- e. Recommendations for further improvements

Tentative Agenda

Review of Alaska Sablefish Stock Assessment

**Alaska Fisheries Science Center
Auke Bay Laboratories
Ted Stevens Marine Research Institute
17109 Pt. Lena Loop Rd.
Juneau, Alaska**

May 10 –12, 2016

Contacts:

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Additional documents, Dana Hanselman, Dana.Hanselman@noaa.gov, 907-789-6626

Tuesday, May 10:

9:00 AM – 10:30 AM: **Introduction**

Topics:

Introductions, adoption of the agenda (and it's relation to TOR), industry concerns, overview of sablefish biology, fishery, history of assessment, prior CIE

10:30 AM – Break

10:45 AM – **Input data (TOR a)**

Topics:

Survey data – abundance indices, ages, lengths, growth, ageing error

Fishery data – abundance indices, ages, lengths, logbooks and observer data

12:00 PM – Lunch

1:00 PM -3:00 PM: **Current Assessment model (TOR b)**

Topics:

Model structure, likelihood formulations, data weighting

3:00 PM – Break

3:15 PM – **Discussions**

5:00 PM – Adjourn for day

Wednesday, May 11:

9:00 AM – 10:30 AM: **Spatial issues (TOR c)**

Topics:

Areal apportionment of catch, movement, and spatially explicit models

10:30 AM – Break

10:45 AM – **Discussions**

12:00 PM – Lunch

1:00 PM – 3:00 PM: **Whale depredation (TOR d)**

Topics:

Estimates of depredation on the survey, fishery, and the effects on assessment

3:00 PM – Break

3:15 PM – Discussions

5:00 PM – Adjourn for day

Thursday, May 12:

9:00 AM -10:30 AM: **Recruitment, ecosystem considerations, future work (TOR e)**

Topics:

Ecosystem considerations, recruitment research, others as requested

10:30 AM – Break

10:45 AM – Discussions

12:00 PM – Lunch

1:00 PM -3:00 PM: **Alternative model runs, further discussion as needed**

3:00 PM – Break

3:15 PM – Further discussions and summarize

5:00 PM – Adjourn meeting